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LCBI Smart Disposition

for LOH due to high K-lot

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W11'25

A3 – LCBI Smart Disposition

BACKGROUND/ BUSINESS CASE:

- At LCBI, any lot having 1A K-lot higher than the SBL limit will be hold without retest 2A.
- Based on LOH trigger, ES/QES will pull the data, pivot to analyze the top pareto bins, define hardware CA and make the disposition (release/retest)
- If disposition is retest, whole lot will be retest instead of retest K-lot only
- In 2024, there are 400+ lots on hold due to high K-lot

PROBLEM STATEMENT:

- The disposition mostly is qualitative decision and human-variance
- Some lots should be retest but disposed as release and vice versa → high K-lot
- Average lot on hold time is 8 hours for 200+ lots
- Additional tool time for retest is 528 hours/year

CURRENT SITUATION:

- The employee look at the data with the quantity of fail units pivot by slot and socket, then assume whether these failed units come from HW or not.
- If "mostly" come from HW then the decision is retest to recover, and if there is no CA which mean fail by unit itself then release the lot.
- Current limit mostly is 15% for all products and rarely review from either PE or ME

CAUSE ANALYSIS

- No standard or limit to evaluate whether this is CA on HW or not.
- No data to prove that the k-lot after retest will be better.
- SBL limit is too tight (15% for 1A test) while recovery rate in 2A is predicted high

IDEAL STATE:

- Auto dispose the lot with optimized result to retest/release without hold

TARGET CONDITION

- Auto calculate the CA percentage of each functional bin
- With the AI model trained by historical data, predict the recovery rate if retest and suggest the optimized disposition

SOLUTION

- For each product, each lot, calculate the fail rate, recovery rate and CA percentage for each of bad functional bin.
- Use last 90 days history data to train the AI model
- When any LOH trigger, query, pre-process data and use pre-trained AI model to predict the recovery rate
- If recovered k-lot > 10% then retest else release

RESULT

- Customer First: Reduce lot on hold time
- Efficiency: Reduce wasting tool time for retest
- Quality: Eliminate human variance or mistakes when handling data and suggest the best disposition

IMPLEMENTATION/STANDARDIZATION:

- Pilot running – Q2'25
- Official release and 100% follow suggestion disposition – Q3'25

K-lot: fail at burn-in
CA: common analysis

Conclusion – Help needed

- Due to current RFC for LOH due to high k-lot (VNAT and VF), there is no quality concern or high-risk bin action needed for lot on hold
- The SBL limit is defined by product level (.bcf file – KlotSBL)
- Most of LOH having high recovery rate, suggest
  - Put the hold trigger after retest 2A
  - Loosen the SBL limit

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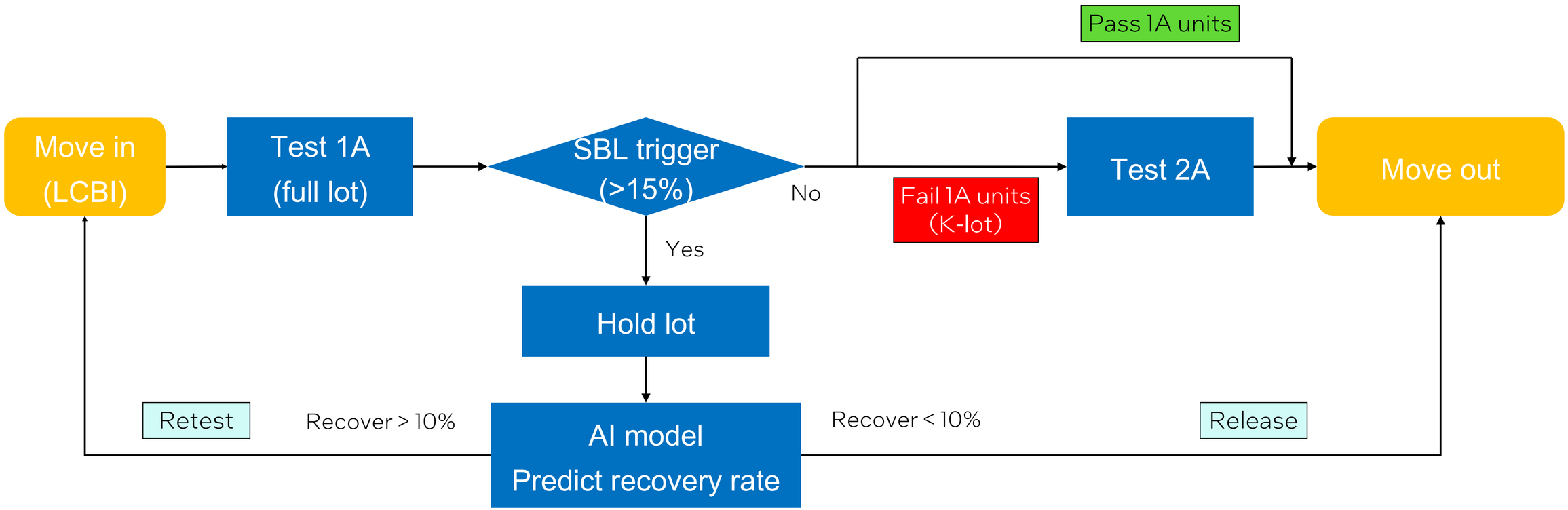
Thank you.

SBL trigger at LCBI



If disposition is retest, whole lot will be retest instead of retest K-lot only

Example of Bin Analysis for HW CA

Count of Bin	DUT																	
Bin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Grand Total
3128																			
1				2						2						2			6
2																1			1
5																1			1
11												3							3
12													3	3					6
13		3							1										4
14	3																		3
15	2	2	2	2	2	2	2	2	2	2	2	2	2		2	2	2	2	34
17	2	2	2				2		2				2	2	2		2	2	20
18										1	1	1							3
21						3													3
28			1																1
6415																			
1																	1	1	2
15													1	2					3
26							1	1											2
9765																			
29						1													1
9984																			
1	2	2	2		2	2	2	2	2		2	2	2	2	2		1	1	28
6	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	36
9981																			
17				2	2	2				2	2	2				2			14

3128																	
3				1													1
4								3			3	3	2				11
6	3	2	1									2					8
8								1		2			2				5
10			5														5
11	4								3								7
12		2															2
13									3	4							7
14				1		2				2		2					7
17					1				1								2
18	2		1	1			1				1		1		2	2	11
20														1		3	4
21					2	3	1					1				1	8
22								2									2
23												2					2
24		3			4	3										1	11
26											1						1
28		1				1						1	1	1	2	1	8
29															2		2
30															2	2	4
6415																	
4		2		2									1	2	2	2	11
5	2		2		1							1					6
6			1			2		2	2			1					8
10			1											2			3
11		2															2
12									2				2				4
13										1							1
17													1				1
18		2	1	1	2	2	1	2		2		2				1	16
20																1	1
21					1				2	1							4

2831																	
1			5														5
2								1									1
3							7										7
4	2																2
5			2				2	3					3	1	1		12
6			5								3	2					10
7													1				1
8		4									2						6
10									2								2
11															4	2	6
12			4			1	2	4									11
13		4						1						1			6
14	1																1
18	1																1
20			2					1				1					4
21										2		1			1		4
22	4	4			1												9
24										1							1
25							2			2				1			5
28														1			1
29	2	5	5	3	2	3	5			1							26
30						4		4								3	11
31													8		2		10
9970																	
9	2	2		2		2	2		2		2					2	16
5328																	
25														1			1
9150																	
2	1		1	1	1	1	1	1		1	1			1	1	1	12
4	1		1		1	1	1	1	1		1						8
5		1		1				1			1			1	1		6
6	1		1	1	1		1				1	1		1	1	1	10
9	1	1		1		1	1		1		1					1	8
13	1	1		1		1		1	1	1	1	1	1	1		1	12
20	1		1	1			1	1			1	1		1		1	9
21			1		1	1	1		1	1	1	1			1	1	10
24	1			1			1		1	1	1			1	1	1	9

Example of Bin Analysis for HW CA

- **Bin summary & CA**

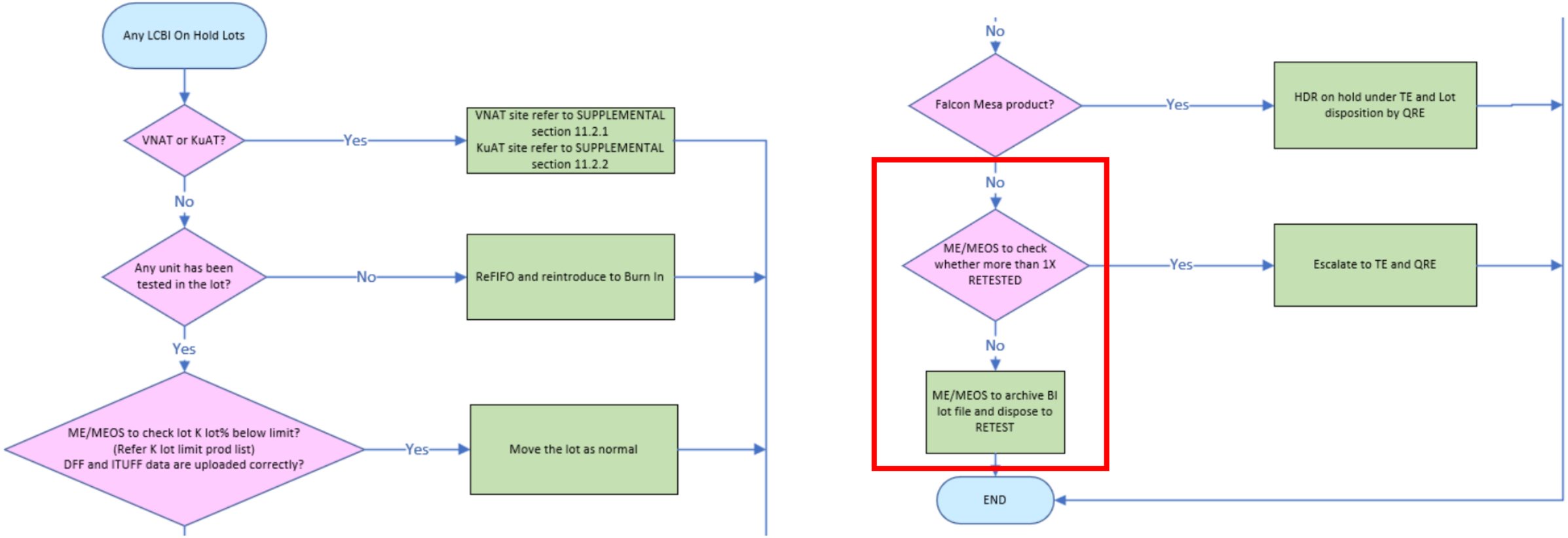
Bin	Count of Bin
100	786
121	4
122	4
131	28
132	9
133	2
804	21
809	1
827	3
837	99
838	7
3128	3
6415	1
9765	4
9766	11
9982	12
9983	1
9984	5
Grand Total	1001

Count of Bin	DUT												
Bin	1	2	3	4	5	6	13	14	15	16	17	18	Grand Total
804													
4		1	1	1	1	1		1					6
5	1				1								2
17	1	1	1					1		1			5
24				1			1			1			3
29	1	1		1		1						1	5
827													
4												2	2
22	1												1
837													
1					1	1	1	1	1				5
3	1	1				1			1				4
4		1		1			1	1		2	2		8
5								1	1			1	3
6				1				1		1	1		4
7		1	1	2	1	2				1	1		9
8					1								1
11						1							1
15										2			2
16		1	1		1				1	1	1		6
17	1	1		1	1		2		3	1		1	11
20	1		1		1			1		1	1	1	7
22		1						1	1		1		4
23		2	1				1		1		1		6
24			2	1					2	1			6
25					1				2	1		1	5
27											1	1	2
28		1				2							3
30		1	1				1		2	3			8
31		1	2									1	4
3128													
15				1									1
22							1						1
30					1								1
6415													
18		1											1
9765													
4							1						1
16							1						1
29								1			1		2
9984													
4			1										1
12						1							1
17				1						1			2
24						1							1
9982													
12	1	1	1	1	1	1	1	1	1	1	1	1	12
9983													
21					1								1
809													
25												1	1
9766													
4						1							1
5			1										1
16												1	1
17						2		1				1	4
24						1		2					3
29										1			1
838													
4					1				1				2
5		1											1
24		2											2
25						1							1
29					1								1

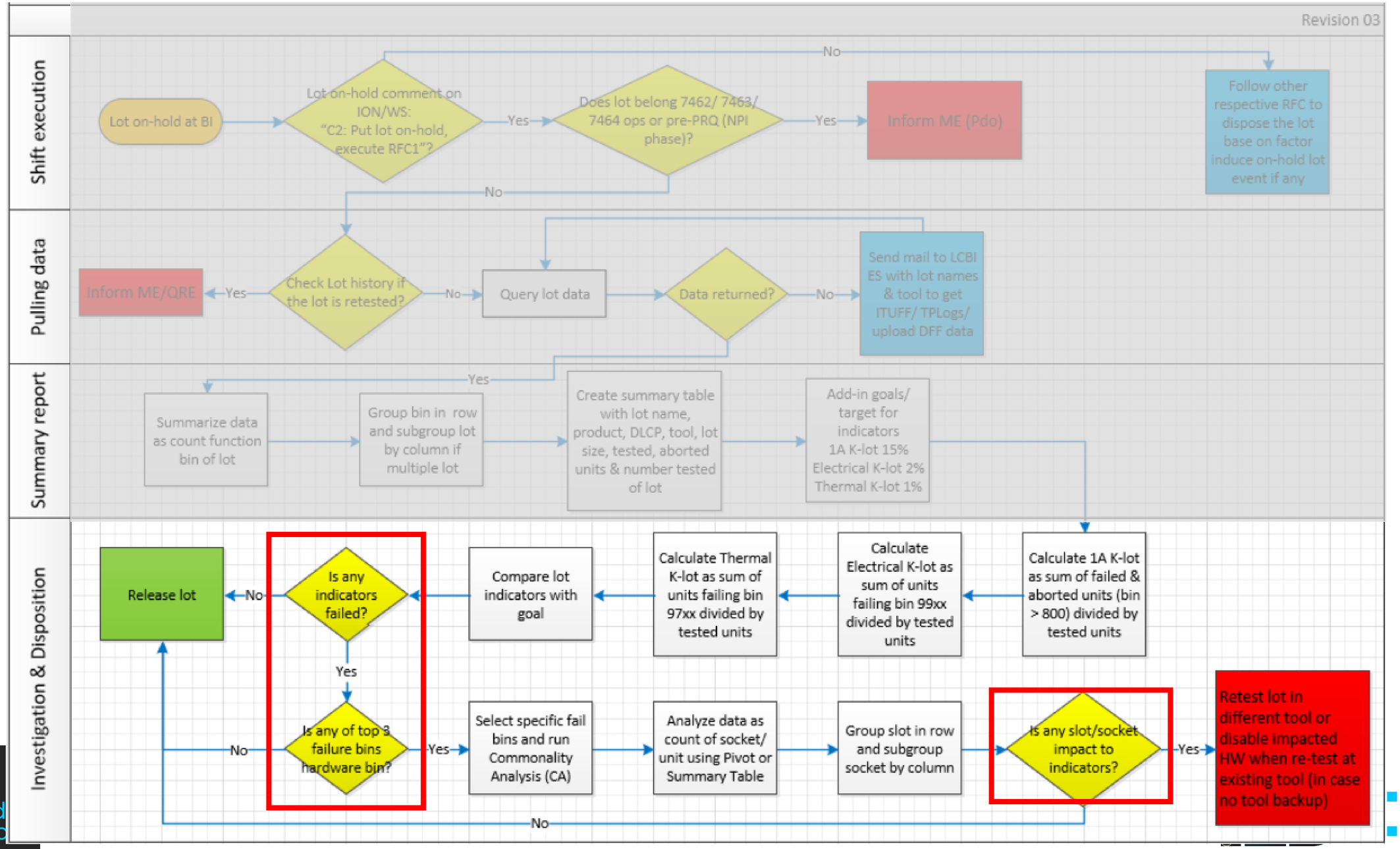
- No CA on any HW but recovery rate is still very high > 80% because of nature of bin 837 for this product only
- More than 20 products with 9000+ kind of bin → couldn't remember or assess the recovery rate base on CA HW only

RFC for LOH due to high K-lot (121-0060)

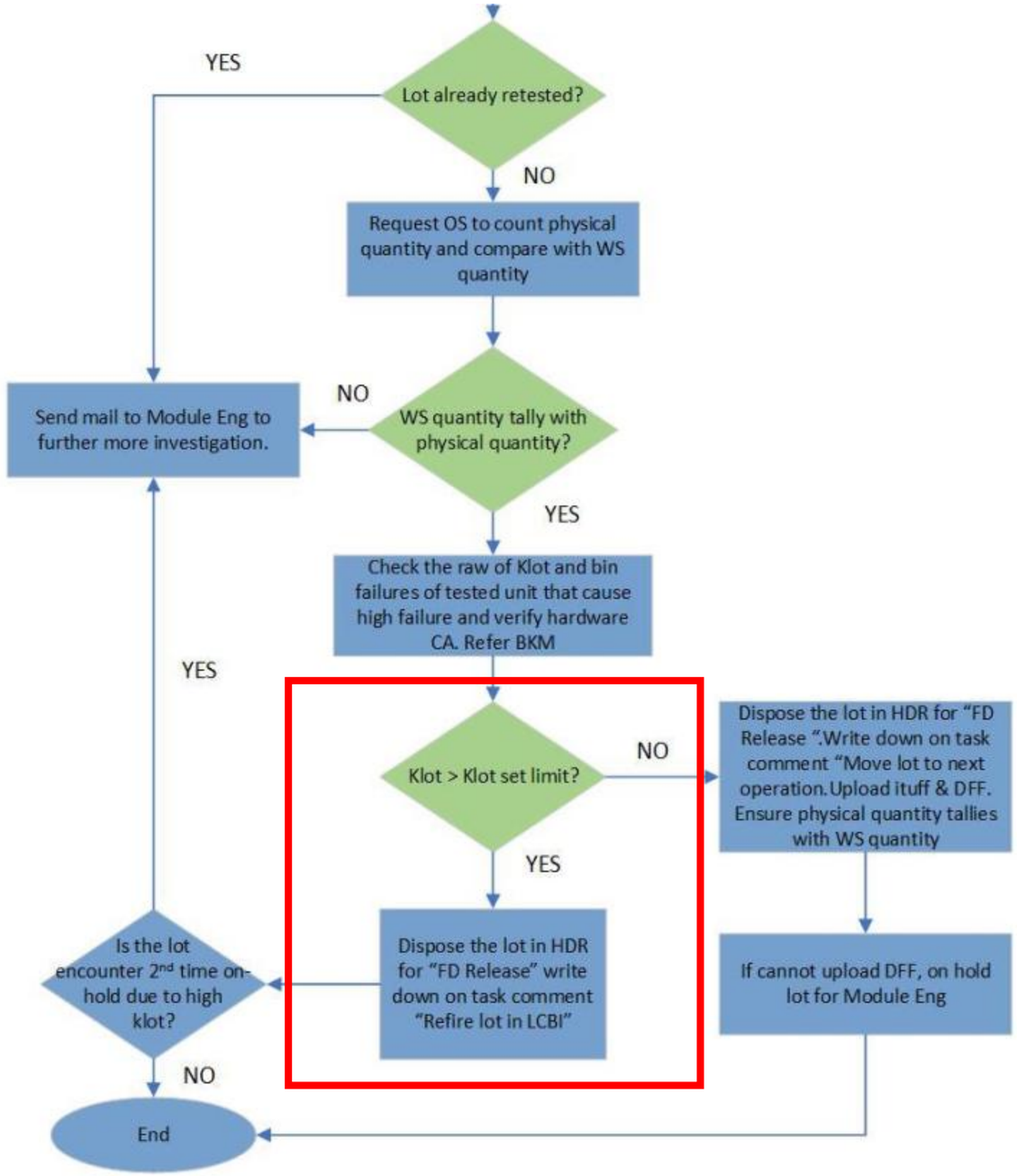
8.1.14 RFC for On Hold Lot Disposition



VNAT RFC


Revision 03
Shift execution
Pulling data
Summary report
Investigation & Disposition
Lot on-hold at BI
Lot on-hold comment on ION/WS: "C2: Put lot on-hold, execute RFC1"?
Yes
No
Does lot belong 7462/ 7463/ 7464 ops or pre-PRQ (NPI phase)?
Yes
No
Inform ME (Pdo)
Follow other respective RFC to dispose the lot base on factor induce on-hold lot event if any
Check Lot history if the lot is retested?
Yes
Inform ME/QRE
No
Query lot data
Data returned?
No
Send mail to LCBI ES with lot names & tool to get ITUFF/ TPLogs/ upload DFF data
Yes
Summarize data as count function bin of lot
Group bin in row and subgroup lot by column if multiple lot
Create summary table with lot name, product, DLCP, tool, lot size, tested, aborted units & number tested of lot
Add-in goals/ target for indicators 1A K-lot 15% Electrical K-lot 2% Thermal K-lot 1%
Calculate 1A K-lot as sum of failed & aborted units (bin > 800) divided by tested units
Calculate Electrical K-lot as sum of units failing bin 99xx divided by tested units
Calculate Thermal K-lot as sum of units failing bin 97xx divided by tested units
Compare lot indicators with goal
Is any indicators failed?
No
Release lot
Yes
Is any of top 3 failure bins hardware bin?
No
Yes
Select specific fail bins and run Commonality Analysis (CA)
Analyze data as count of socket/ unit using Pivot or Summary Table
Group slot in row and subgroup socket by column
Is any slot/socket impact to indicators?
Yes
Retest lot in different tool or disable impacted HW when re-test at existing tool (in case no tool backup)
No

KuAT RFC


Lot already retested?
YES
NO
Request OS to count physical quantity and compare with WS quantity
WS quantity tally with physical quantity?
NO
Send mail to Module Eng to further more investigation.
YES
Check the raw of Klot and bin failures of tested unit that cause high failure and verify hardware CA. Refer BKM
YES
Klot > Klot set limit?
NO
Dispose the lot in HDR for "FD Release ".Write down on task comment "Move lot to next operation.Upload ituff & DFF. Ensure physical quantity tallies with WS quantity
YES
Dispose the lot in HDR for "FD Release" write down on task comment "Refire lot in LCBI"
If cannot upload DFF, on hold lot for Module Eng
Is the lot encounter 2nd time on-hold due to high klot?
NO
End

Details of LOH

Total trigger of LOH last 360 days

Row Labels	#lot	volume	Average of Klot	Sum of time_on_hold	Sum of testing_time
MVOU	**91**	**134063**	**13.73**	**29.9**	**23.5**
RPR8161	50	77798	11.87	11.9	4.7
RPL8161	24	36374	18.93	11.7	17.3
KBZ	5	8071	13.38	0.7	0.8
SKA	3	3212	10.95	0.2	0.1
CMLS62	2	2938		0.2	0.2
CFD	2	3444	11.85	0.1	0.2
RPLPS6	1	96	13.79	0.6	0.0
CFA	1	224		0.6	0.0
CMLS102	1	210	27.62	3.7	0.0
GRR	1	130		0.2	0.1
KBLU22	1	1566		0.1	0.0
TRST	**113**	**167896**	**2.99**	**32.7**	**22.3**
RPR8161	46	71944	1.26	12.4	5.2
RPL8161	15	23436	1.34	5.5	9.1
WHLU42	11	17316	4.70	1.9	0.9
CMLS102	10	16916	9.21	1.7	1.7
KBLU22	9	11041		0.9	1.1
RPLP682	9	8698	0.00	2.3	0.7
RRC8161	4	6272		3.0	2.6
CFD	3	5161	2.27	0.9	0.3
CFA	2	1946		0.1	0.1
RPG8161	1	1568		0.6	0.3
KBZ	1	1722	3.31	0.3	0.1
RPLPS6	1	84	0.00	0.2	0.0
RPRP682	1	1792		2.9	0.2
Grand Total	**204**	**301959**	**7.38**	**62.6**	**45.8**

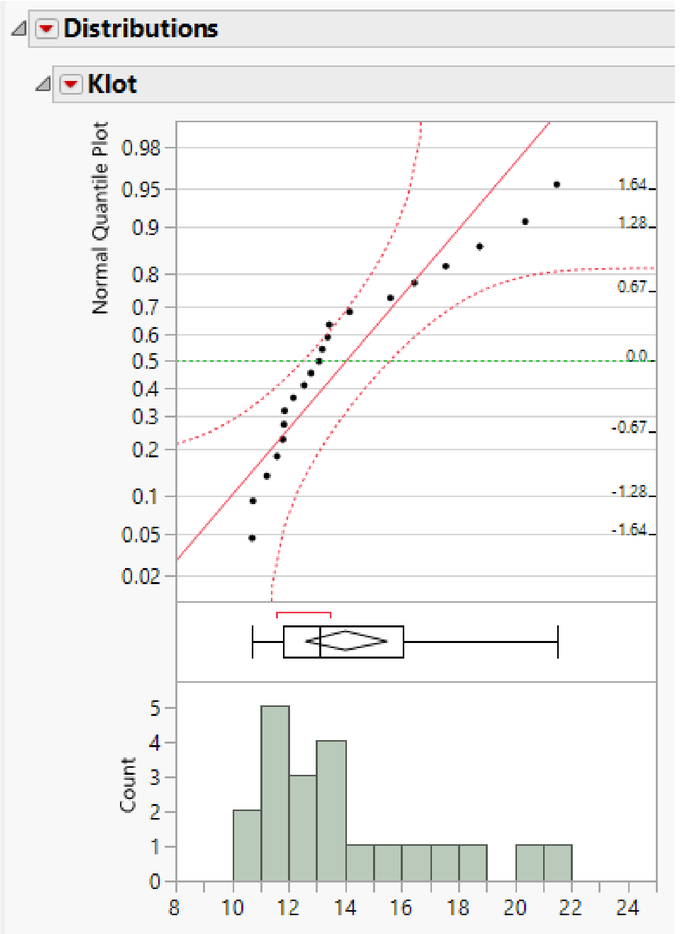
Recovery rate base on last 180 days historical data

	Release		Retest		
Row Labels	Count of lot	Average of Klot	Count of lot	Average of Klot	Recovery_rate
RPR8161	50	11.87	46	1.26	**83%**
RPL8161	24	18.93	15	1.34	**78%**
WHLU42			11	4.70	**87%**
CMLS102	1	27.62	10	9.21	**67%**
KBLU22	1		9		**83%**
RPLP682			9	0.00	**92%**
KBZ	5	13.38	1	3.31	**87%**
CFD	2	11.85	3	2.27	**60%**
RRC8161			4		**No Data**
SKA	3	10.95			**90%**
CFA	1		2		**45%**
RPLPS6	1	13.79	1	0.00	**No Data**
CMLS62	2				**89%**
GRR	1				**84%**
RPRP682			1		**91%**
RPG8161			1		**93%**

Most of products have recovery rate higher than 80% at 2A

RPL8161 trigger on hold high klot

Disposition: release (no retest)

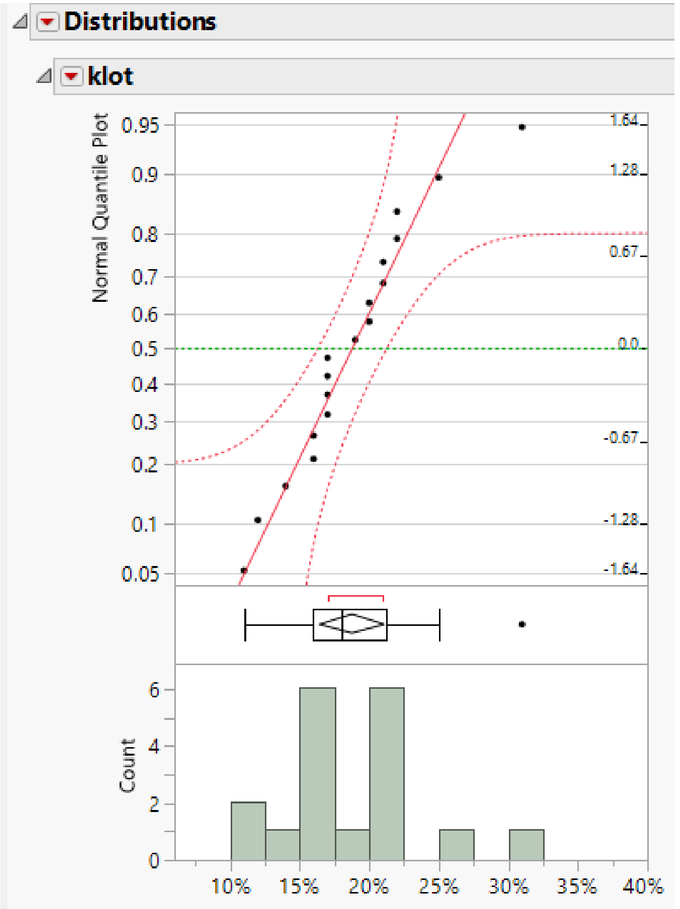


lot	1A Klot	recovery_rate_predicted	final_klot_predicted
U408P820	12.79	10.45	2.34
U440D822	21.49	18.23	3.26
U440D844	20.36	17.78	2.58
U441D936	16.44	13.82	2.62
U442F541	18.76	15.99	2.77
U444B549	11.58	8.74	2.84
U444D146	11.87	9.66	2.21
U446D884	11.79	9.36	2.43
U446F545	14.15	11.36	2.79
U446G255	10.71	7.91	2.80
U449H304	11.20	8.79	2.41
U450H154	12.18	9.55	2.63
U451C063	13.38	11.04	2.34
U451G625	17.58	15.69	1.89
U503B649	10.74	7.59	3.15
U503B672	13.01	9.95	3.06
U503C020	15.61	11.88	3.73
U504C113	13.44	9.69	3.75
U504C514	11.83	8.58	3.25
U504D419	13.18	9.02	4.16
U507C047	12.57	8.64	3.93
		233.72	

- All of lots is predicted to have high recovery rate
- If all lots retest 2A, the predicted k-lot units could be saved is 233% x 1.4 ku = **3.2 ku**

RPL8161 trigger on hold high klot

Disposition: retest full lot



lot	1st	2nd 1A	2nd 2A	3rd 1A	3rd 2A
U443F146	17%	7%	2%		
U444B226	22%	1%	0%		
U444B937	14%	2%	0%		
U444C704	20%	4%	0%		
U446H164	11%	1%	0%		
U450H021	12%	2%	0%		
U450H461	19%	5%	1%		
U451G504	20%	5%	1%		
U451G676	16%	12%	2%		
U501C876	21%	6%	2%		
U501C964	17%	6%	1%		
U502C609	17%	2%	0%		
U502C788	17%	4%	0%		
U502G684	22%	25%		10%	2%
U503B672	21%	13%			
U504C474	16%	4%	0%		
U507C738	31%	1%	0%		

- Most of lots after retest have high recovery rate and final K-lot around 1%
- Some lots still observe high k-lot after retest because when full lot retest, there is percentage that pass units in 1st test switch to failed units in 2nd test

→ Loosen the SBL limit so that these lots could be retest 2A (retest k-lot only) without hold and retest full lot

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Thank you.